

Creation Date 20-Oct-2009

Revision Date 04-Oct-2023

Revision Number 11

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                       |
|----------------------------------|-------------------------------------------------------|
| <b>Product Description:</b>      | <u>Iodine</u>                                         |
| <b>Cat No. :</b>                 | 387050000; 387050010; 387050250; 387051000; 387052500 |
| <b>Index No</b>                  | 053-001-00-3                                          |
| <b>CAS No</b>                    | 7553-56-2                                             |
| <b>EC No</b>                     | 231-442-4                                             |
| <b>Molecular Formula</b>         | I <sub>2</sub>                                        |
| <b>REACH registration number</b> | 01-2119485285-30                                      |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor - Fisher Scientific AG**  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

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## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Acute oral toxicity                                  | Category 4 (H302) |
| Acute dermal toxicity                                | Category 4 (H312) |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 4 (H332) |
| Skin Corrosion/Irritation                            | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319) |
| Specific target organ toxicity - (single exposure)   | Category 3 (H335) |
| Specific target organ toxicity - (repeated exposure) | Category 1 (H372) |

##### Environmental hazards

|                        |                   |
|------------------------|-------------------|
| Acute aquatic toxicity | Category 1 (H400) |
|------------------------|-------------------|

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H372 - Causes damage to organs through prolonged or repeated exposure  
Thyroid  
H400 - Very toxic to aquatic life

#### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell

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## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment  
Lachrymator (substance which increases the flow of tears)  
Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component | CAS No    | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                  |
|-----------|-----------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iodine    | 7553-56-2 | 231-442-4 | >95      | Acute Tox. 4 (H302)<br>Acute Tox. 4 (H312)<br>Acute Tox. 4 (H332)<br>Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)<br>STOT RE 1 (H372)<br>Aquatic Acute 1 (H400) |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Iodine    | -                                     | 1        | -               |

|                           |                  |
|---------------------------|------------------|
| REACH registration number | 01-2119485285-30 |
|---------------------------|------------------|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

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## Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes. Do not allow run-off from fire-fighting to enter drains or water courses.

#### **Hazardous Combustion Products**

Hydrogen iodide.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

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## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store in metal containers. Keep at temperatures below 25°C.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 6.1D

Switzerland - Storage of hazardous substances

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union | The United Kingdom                                         | France                                                      | Belgium                                                                                                                       | Spain                                                                                                                                                                         |
|-----------|----------------|------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iodine    |                | STEL: 0.1 ppm 15 min<br>STEL: 1.1 mg/m <sup>3</sup> 15 min | STEL / VLCT: 0.1 ppm.<br>STEL / VLCT: 1 mg/m <sup>3</sup> . | TWA: 0.01 ppm 8 uren<br>TWA: 0.1 mg/m <sup>3</sup> 8 uren<br>STEL: 0.1 ppm 15 minuten<br>STEL: 1 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 0.1 ppm (15 minutos).<br>STEL / VLA-EC: 1 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 0.01 ppm (8 horas)<br>TWA / VLA-ED: 0.1 mg/m <sup>3</sup> (8 horas) |

| Component | Italy | Germany | Portugal                                          | The Netherlands | Finland                                                                           |
|-----------|-------|---------|---------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|
| Iodine    |       | Haut    | STEL: 0.1 ppm 15 minutos<br>TWA: 0.01 ppm 8 horas |                 | STEL: 0.1 ppm 15 minuutteina<br>STEL: 1.1 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component | Austria                                                                                                                                                                                                      | Denmark                                          | Switzerland                                                                                                                                   | Poland                                                                          | Norway                                           |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------|
| Iodine    | Haut<br>MAK-KZGW: 0.1 ppm 15 Minuten<br>MAK-KZGW: 1 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.1 ppm 8 Stunden<br>MAK-TMW: 1 mg/m <sup>3</sup> 8 Stunden<br>Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> | Haut/Peau<br>STEL: 0.1 ppm 15 Minuten<br>STEL: 1 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.1 ppm 8 Stunden<br>TWA: 1 mg/m <sup>3</sup> 8 Stunden | STEL: 1 mg/m <sup>3</sup> 15 minutach<br>TWA: 0.5 mg/m <sup>3</sup> 8 godzinach | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> |

| Component | Bulgaria                   | Croatia                                                                          | Ireland                                                                                                        | Cyprus | Czech Republic                                                         |
|-----------|----------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------|
| Iodine    | TWA: 3.0 mg/m <sup>3</sup> | STEL-KGVI: 0.1 ppm 15 minutama.<br>STEL-KGVI: 1.1 mg/m <sup>3</sup> 15 minutama. | TWA: 0.01 ppm 8 hr. inhalable fraction and vapour<br>TWA: 0.01 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.1 ppm 15 min |        | TWA: 0.1 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 1 mg/m <sup>3</sup> |

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| Component | Estonia                                                            | Gibraltar | Greece                                                                                 | Hungary                                                                                                                        | Iceland                                    |
|-----------|--------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Iodine    | STEL: 0.1 ppm 15 minutes.<br>STEL: 1 mg/m <sup>3</sup> 15 minutes. |           | STEL: 0.1 ppm<br>STEL: 1 mg/m <sup>3</sup><br>TWA: 0.1 ppm<br>TWA: 1 mg/m <sup>3</sup> | STEL: 1 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 1 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borön keresztüli felszívódás | STEL: 0.1 ppm<br>STEL: 1 mg/m <sup>3</sup> |

| Component | Latvia                   | Lithuania                                        | Luxembourg | Malta | Romania                                                                                                                   |
|-----------|--------------------------|--------------------------------------------------|------------|-------|---------------------------------------------------------------------------------------------------------------------------|
| Iodine    | TWA: 1 mg/m <sup>3</sup> | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> |            |       | TWA: 0.09 ppm 8 ore<br>TWA: 0.5 mg/m <sup>3</sup> 8 ore<br>STEL: 0.2 ppm 15 minute<br>STEL: 1 mg/m <sup>3</sup> 15 minute |

| Component | Russia                                    | Slovak Republic                                                              | Slovenia | Sweden                                                                           | Turkey |
|-----------|-------------------------------------------|------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------|--------|
| Iodine    | Skin notation<br>MAC: 1 mg/m <sup>3</sup> | Ceiling: 1.1 mg/m <sup>3</sup><br>TWA: 0.1 ppm<br>TWA: 1.1 mg/m <sup>3</sup> |          | Binding STEL: 0.1 ppm 15 minuter<br>Binding STEL: 1 mg/m <sup>3</sup> 15 minuter |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                   | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-----------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Iodine<br>7553-56-2 ( >95 ) |                              |                                 |                                | DNEL = 0.01mg/kg bw/day           |

| Component                   | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-----------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Iodine<br>7553-56-2 ( >95 ) |                                  |                                     |                                    | DNEL = 0.07mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                   | Fresh water      | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|-----------------------------|------------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Iodine<br>7553-56-2 ( >95 ) | PNEC = 18.13µg/L | PNEC = 3.99mg/kg sediment dw |                    | PNEC = 11mg/L                      | PNEC = 5.95mg/kg soil dw |

| Component | Marine water | Marine water sediment | Marine water Intermittent | Food chain | Air |
|-----------|--------------|-----------------------|---------------------------|------------|-----|
|           |              |                       |                           |            |     |

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|                             |                  |                                     |  |  |  |
|-----------------------------|------------------|-------------------------------------|--|--|--|
| Iodine<br>7553-56-2 ( >95 ) | PNEC = 60.01µg/L | PNEC =<br>20.22mg/kg<br>sediment dw |  |  |  |
|-----------------------------|------------------|-------------------------------------|--|--|--|

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                 |                          |            |
|---------------------------------|--------------------------|------------|
| <b>Physical State</b>           | Solid                    |            |
| <b>Appearance</b>               | Grey                     |            |
| <b>Odor</b>                     | pungent                  |            |
| <b>Odor Threshold</b>           | No data available        |            |
| <b>Melting Point/Range</b>      | 113 °C / 235.4 °F        |            |
| <b>Softening Point</b>          | No data available        |            |
| <b>Boiling Point/Range</b>      | 185 °C / 365 °F          | @ 760 mmHg |
| <b>Flammability (liquid)</b>    | Not applicable           | Solid      |
| <b>Flammability (solid,gas)</b> | No information available |            |

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | 5.1                      | saturated solution                |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | 0.3 g/L (20°C)           | practically insoluble             |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Iodine                                  | 2.49                     |                                   |
| Vapor Pressure                          | 0.41 hPa @ 25 °C         |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | ~ 2100 kg/m <sup>3</sup> |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

## 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | I <sub>2</sub>         |
| Molecular Weight  | 253.81                 |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Avoid dust formation. Incompatible products. Excess heat.

### 10.5. Incompatible materials

Strong oxidizing agents. Finely powdered metals. Ammonia. Alcohols. copper.

### 10.6. Hazardous decomposition products

Hydrogen iodide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 4 |
| Dermal              | Category 4 |
| Inhalation          | Category 4 |

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| Component | LD50 Oral         | LD50 Dermal           | LC50 Inhalation       |
|-----------|-------------------|-----------------------|-----------------------|
| Iodine    | 315 mg/kg ( Rat ) | 1425 mg/kg ( Rabbit ) | 4.588 mg/L 4h ( Rat ) |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

| Component                   | Test method                                       | Test species | Study result    |
|-----------------------------|---------------------------------------------------|--------------|-----------------|
| Iodine<br>7553-56-2 ( >95 ) | OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs

Respiratory system.

(i) STOT-repeated exposure; Category 1

Target Organs

Thyroid.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish      | Water Flea           | Freshwater Algae     |
|-----------|----------------------|----------------------|----------------------|
| Iodine    | LC50 = 1.67 mg/L 96h | EC50 = 0.55 mg/L 48h | EC50 = 0.13 mg/L 72h |

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| Component | Microtox           | M-Factor |
|-----------|--------------------|----------|
| Iodine    | EC50 = 280 mg/L 3h | 1        |

## 12.2. Persistence and degradability

|                                       |                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Persistence                           | Persistence is unlikely.                                                                                        |
| Degradability                         | Not relevant for inorganic substances.                                                                          |
| Degradation in sewage treatment plant | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Iodine    | 2.49    | No data available             |

## 12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

## 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN3495

ACR38705

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**14.2. UN proper shipping name** IODINE  
**14.3. Transport hazard class(es)** 8  
**Subsidiary Hazard Class** 6.1  
**14.4. Packing group** III

## ADR

**14.1. UN number** UN3495  
**14.2. UN proper shipping name** IODINE  
**14.3. Transport hazard class(es)** 8  
**Subsidiary Hazard Class** 6.1  
**14.4. Packing group** III

## IATA

**14.1. UN number** UN3495  
**14.2. UN proper shipping name** IODINE  
**14.3. Transport hazard class(es)** 8  
**Subsidiary Hazard Class** 6.1  
**14.4. Packing group** III

**14.5. Environmental hazards** Dangerous for the environment  
 Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Iodine    | 7553-56-2 | 231-442-4 | -      | -   | X     | X    | KE-21023 | X    | -    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Iodine    | 7553-56-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Iodine    | 7553-56-2 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

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## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Iodine    | 7553-56-2 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------|---------------------------------------|-------------------------|
| Iodine    | WGK2                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                   | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Iodine<br>7553-56-2 ( >95 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed

H312 - Harmful in contact with skin

H332 - Harmful if inhaled

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

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H372 - Causes damage to organs through prolonged or repeated exposure  
H400 - Very toxic to aquatic life

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical incident response training.

**Creation Date** 20-Oct-2009

**Revision Date** 04-Oct-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**